

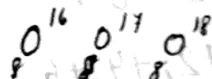
Element	Atomic No.	Group-16 elements	
O	8	$[\text{He}] 2s^2 2p^4$	First four elements
S	16	$[\text{Ne}] 3s^2 3p^4$	- Chalcogens
Se	34	$[\text{Ar}] 3d^{10} 4s^2 4p^4$	Po - Radioactive
Te	52	$[\text{Kr}] 4d^{10} 5s^2 5p^4$	shortest half life = 13.8 days
Po	84	$[\text{Xe}] 4f^{14} 5d^{10} 6s^2 6p^4$	Lv - synthetic
Lv	116	$[\text{Rn}] 7s^2 7p^4$	Radioactive

Po → decays as
Th, U

O is most abundant 46.6% by mass on Earth crust.

Dry air - 20.946% of O_2 volume

"O" has 3 isotopes



99.762%, 0.038%, 0.2%

"S" - sulphates & sulphides

Gypsum - $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

Epsom salt - $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$

Barytes - BaSO_4

Galena - PbS

Zinc Blende - ZnS

Copper pyrites - CuFeS_2

→ 'S' is present as H_2S in egg, garlic, onion, hair, wool and protein.

→ Native 'S' is present in volcanic areas.

General Properties

O_2 - Di ($\text{O}=\text{O}$)

$\text{S}_8, \text{Se}_8, \text{Te}_8$ - octa

Po - mono

→ physical state

O_2 - gas

$\text{S}_8, \text{Se}_8, \text{Te}_8, \text{Po}$ - solids

→ covalent radius

O to Te ↑

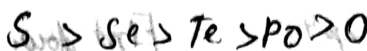
→ I.P. O to Po ↓

→ E.N. O to Po ↓

O more

electronegative element.

→ E. Affinity



'O' has low E.A

Small size and e⁻ repulsion

→ M.P



→ Density



→ B.P



→ O, S - non metals

Se, Te - metalloids

Po - radioactive metals

→ Oxidation state

Common O. state -2 (O. state)

+2, +4, +6 O. state

O -2 (H₂O, MgO, Na₂O)

peroxides -1 (H₂O₂, BaO₂)

superoxides -1/2 (KO₂, RbO₂, CsO₂)

O₂F₂ +1

OF₂ +2

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"O" does not exhibit higher oxidation state due to absence of d-orbitals.

"S"

Ground state - 3s² 3p⁴ -2 (O₂) +2



1st E. state - 3s² 3p³ 3d¹ +4



2nd E. state - 3s¹ 3p³ 3d² +6



→ Higher O. state ability ↓ decreases from top to bottom.

→ "O" - Small size

- high EN

- high I.P

Max. covalency of "O" is 4.

Allotropy

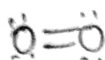
→ O_2 - Two types of allotropic forms

O_2 and O_3

para magnetic

Colourless gas

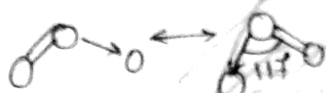
Diatomic



diamagnetic

pale blue gas

Triatomic



Bond length - 128 pm

→ "S" has more no. of allotropic forms (has high catenation capacity in VIA group)

1) Rhombic sulphur (α -sulphur)

2) Monoclinic sulphur (β -sulphur)

3) γ -Monoclinic sulphur

4) Engel sulphur

5) Plastic sulphur

→ Rhombic "S" is most stable form (α -sulphur)

sulphur exists in CS_2 - α sulphur

M.P = 385.8 K S.G (specific gravity) = 2.06 g/cm³

yellow colour. insoluble in water

soluble in non-polar solvent in benzene, alcohol,

ether and CS_2

→ α -sulphur is melted and cooled β -sulphur formed.

β -sulphur

M.P = 392.2 K

S.G = 1.98 g/cm³

→ γ -sulphur

α , β and γ sulphur have S_8 rings

S_8 - puckered ring (8c) crown structure



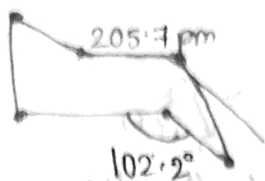
S-S bond length 204 pm

→ Engel's sulphur

$\text{Na}_2\text{S}_2\text{O}_3$ with HCl precipitates



S_8 has chair form



Sulphur heated $\sim 1000^\circ\text{K}$ forms S_2

stable upto 2200°C

S_2 is paramagnetic like O_2

→ Se - 6 allotropic forms

3 - non metallic red colour forms

1 - red colour amorphous forms

2 - Grey colour forms

Grey forms are stable

→ Te - has only one allotropic form

silvery white metallic form

→ Po - 2 forms

α -form (cubic)

β -form (Rhombohedral)

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Hydrides: H_2M type hydrides of Group VIA

H_2O - water

H_2S - hydrogen sulphide

H_2Se - hydrogen selenide

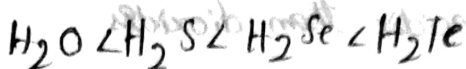
H_2Te - hydrogen telluride

H_2Po - Hydrogen polonide

H_2S , H_2Se , H_2Te colourless, poisonous gases

H_2Po unstable at room temp.

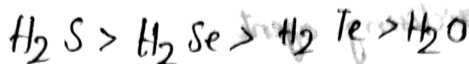
→ Aq. solutions - acidic



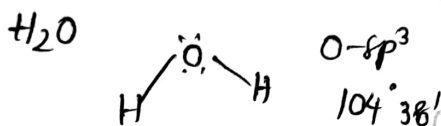
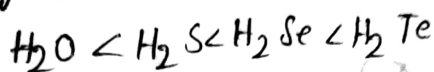
→ B.P order $H_2S < H_2Se < H_2Te < H_2O$

H_2O has abnormally high B.P due to presence of H-bonds

→ Order of volatility

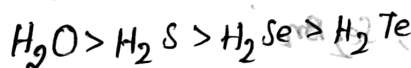


→ Reducing property



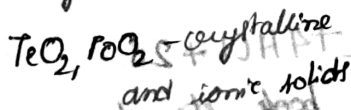
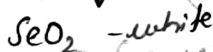
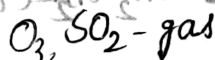
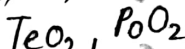
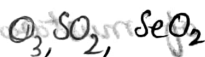
→ In H_2S - 92.3°, H_2Se - 91°, H_2Te - 90° p-orbitals are involved in bond formation

→ Bond angle

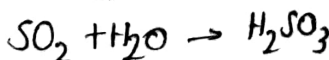
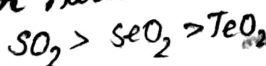


Oxides:

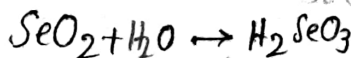
Dioxides



Acidic nature



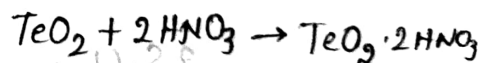
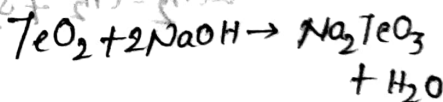
sulphurous acid



selenous acid

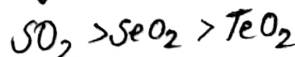
Tetroxides

TeO_2 & PoO_2 are amphoteric



SO_2 - strong reducing agent

Reducing property

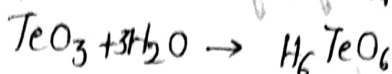
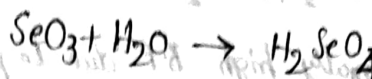
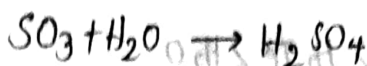


TeO_2 - oxidising agent

Tellurides



Tellurides are more acidic than dioxides



telluric acid

SO_3 - oxidising agent

Halides : "s" forms

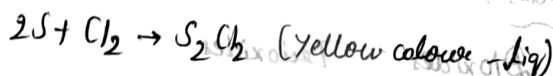
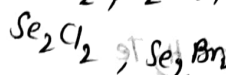
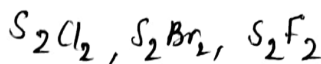
1) Monohalides (M_2X_2)

2) Dihalides (MX_2)

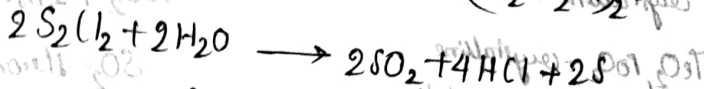
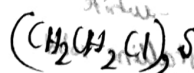
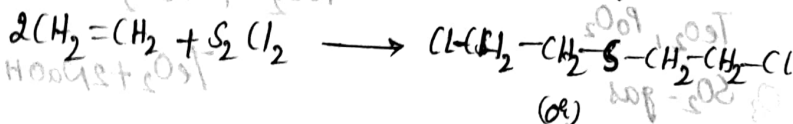
3) Tetrahalides (MX_4)

4) Hexahalides (MX_6)

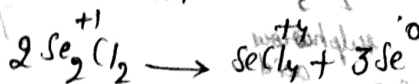
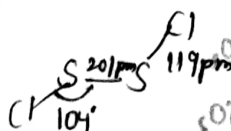
Monohalides :



S_2Cl_2 used for preparation of mustard gas



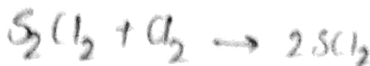
S_2Cl_2 has open book structure



Monohalides undergo disproportionation reaction

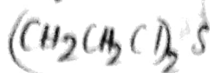
Dihalides: $SF_2, SCl_2, TeCl_2$

"Se" dihalides are unstable



SCl_2 - Red colour, foul smelling liquid

SCl_2 also forms mustard gas with ethylene



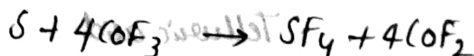
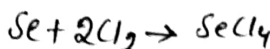
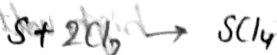
S-sp³
2-1 sp

S

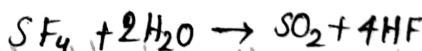
98.3°

Tetrahalides: MX_4

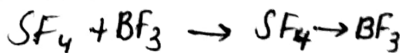
$SF_4, SCl_4, TeCl_4, SeCl_4, SeBr_4$



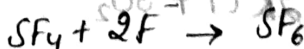
SF_4 - gas, SeF_4 - liquid, TeF_4 - solid



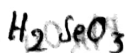
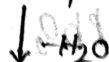
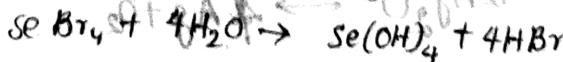
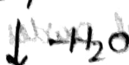
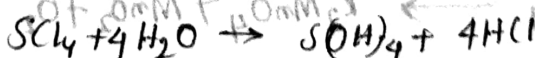
Lewis base

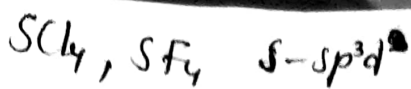


Lewis acid



Good fluorinating agent

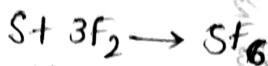
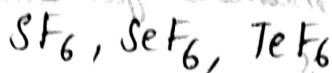




TeCl_4 - more stable

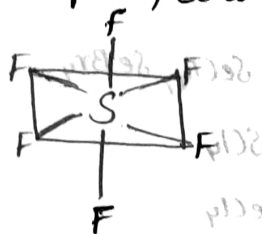
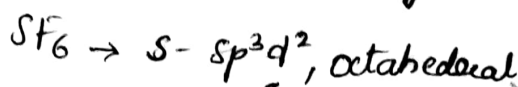


Hexahalides (MX_6):



SF_6 colourless, non-toxic, non-inflammable gas

SF_6 - chemically inert



Used as insulator in

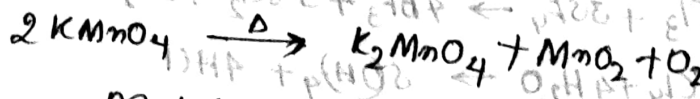
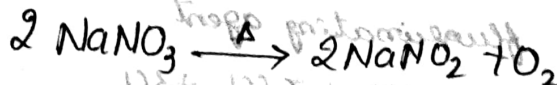
high voltage conductors



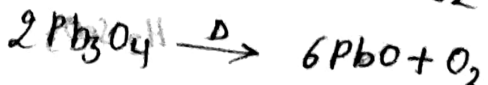
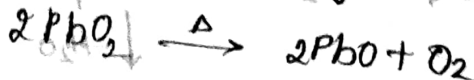
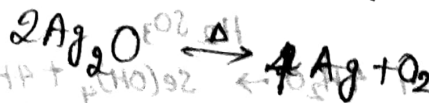
Telluric acid

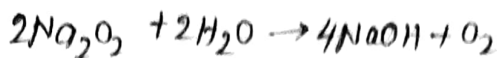
Dioxygen (O_2):

O_2 by thermal decomposition of chlorates, nitrates, permagnates.

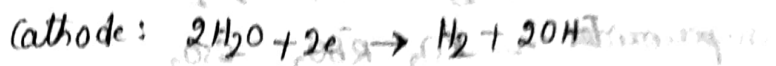
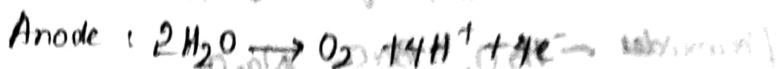


Metal oxides on heating gives O_2

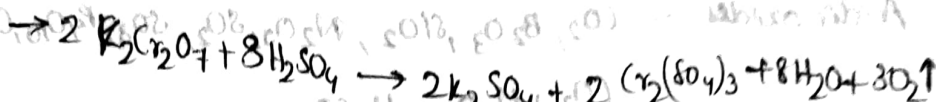
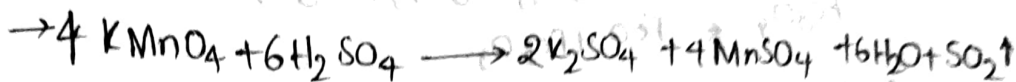
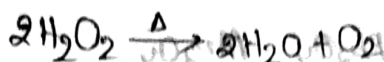




Electrolysis of water gives O_2 at anode



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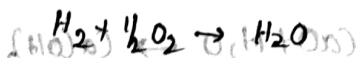
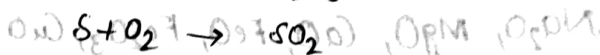
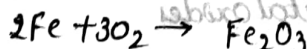
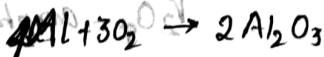
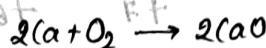
$\rightarrow \text{O}_2$ is colourless gas

liquefies at 90K & freezes at 55K

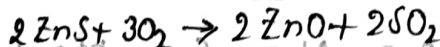
soluble in water

3.08 cm^3 in 100 cm^3 of water

Reacts with metals & non-metals



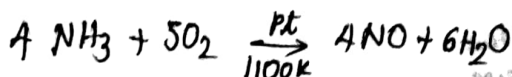
Roasting process



Combustion reaction



Deacon's process



O_2 - used in hospitals, liquid O_2 & N_2H_4 as rocket
hydrogen fuel.

Oxides

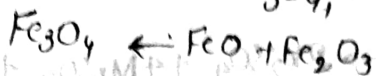
Normal oxides - H_2O , MgO , Al_2O_3 , Na_2O

Peroxides - H_2O_2 , BaO_2 , Na_2O_2

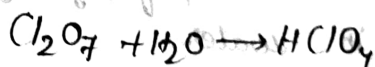
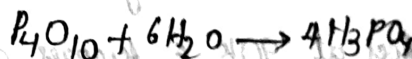
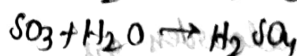
Superoxides - KO_2 , RbO_2 , CSO_2

Suboxides - C_3O_2 , N_2O

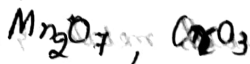
Mixed oxides - Fe_3O_4 , Pb_3O_4 , Mn_3O_4



Acidic oxides - CO_2 , B_2O_3 , SiO_2 , N_2O_5 , SO_2 , SO_3 , P_4O_{10} , Cl_2O_7



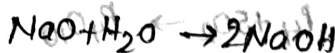
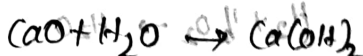
Some of the metal oxides having acidic nature



V_2O_5 - amphoteric acidic oxide

Basic oxides - metal oxides

Na_2O , MgO , CaO , FeO , Fe_2O_3 , CuO



Amphoteric oxides

SiO_2 , ZnO , BeO , Al_2O_3 , SnO_2 , PbO , V_2O_5 , Sb_2O_3

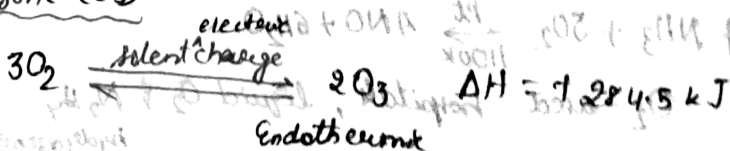
Neutral oxides - N_2O , CO , NO , H_2O

CrO - basic

Cr_2O_3 - amphoteric

CrO_3 - acidic

Ozone (O_3)



Siemann's ozoniser

Beaudie's ozoniser

Mixture $O_2 + O_3$ - Ozonised mixture

O_3 is pale blue gas at boiling
dark blue liquid
violet black solid

O_3 in low concentration upto 100 ppm is harmless
 $O_3 > 100$ ppm causes nausea, headache.



$$\Delta H = -ve$$

$$\Delta S = +ve$$

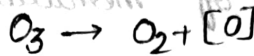
$$\Delta G = \Delta H - T\Delta S$$

$$\Delta G = -ve$$

Conversion of O_3 to O_2
takes place spontaneously

Spontaneous

O_3 - good oxidising agent

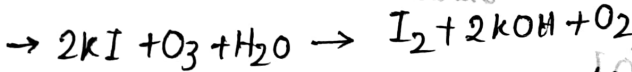


O_3 is weak O-Agent than F_2
strong O-Agent than H_2O_2

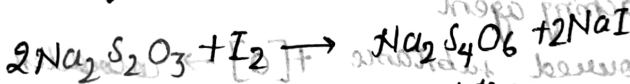


Black

white



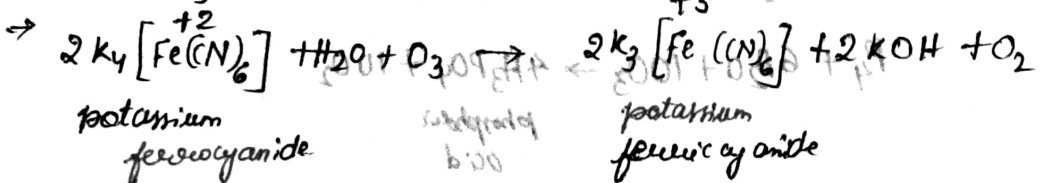
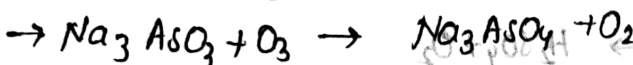
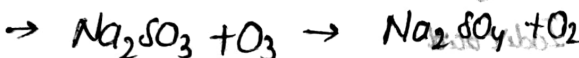
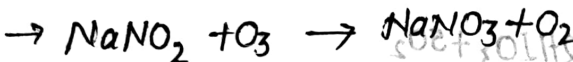
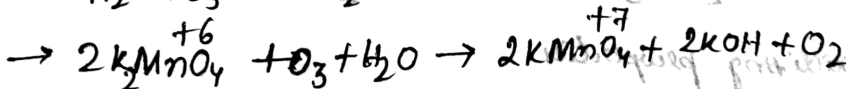
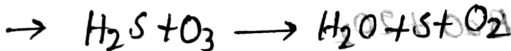
Reaction of is useful for estimation of O_3



Standard

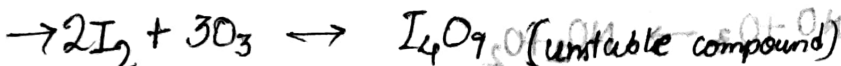
sodium tetrathionate

thionate



potassium
ferrocyanide

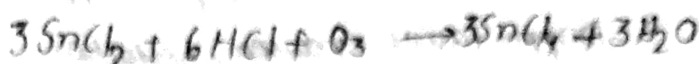
potassium
ferricyanide



* SO_2 is oxidised to SO_3



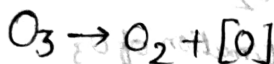
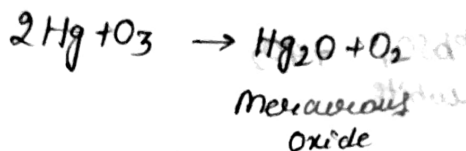
SnCl_2 is oxidised to SnCl_4



Tailing of Mercury

Mercury loses lustrous and meniscus nature in the presence of O_3 and it is stick on glass surface due to the formation of mercurous oxide. This is called tailing of mercury.

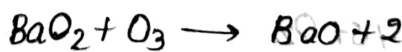
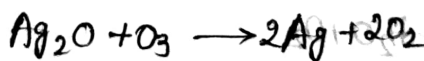
Hg - Lustrous nature
Meniscus nature



Bleaching agent

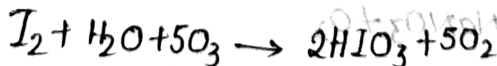
Coloured substance $+ [\text{O}] \rightarrow$ colourless

$\text{O}_3 \rightarrow$ Reducing agent

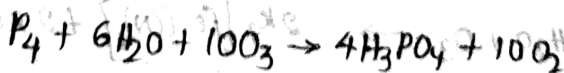


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Oxidising properties



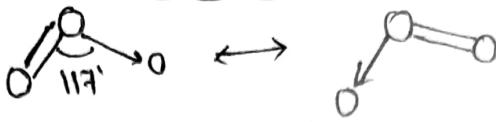
Iodic acid



phosphoric
acid



Structure of O_3



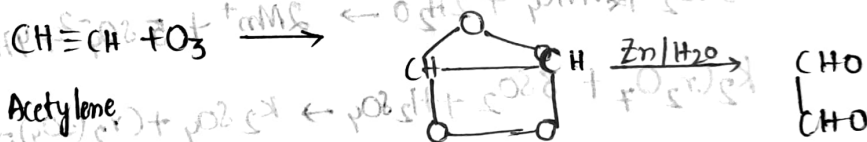
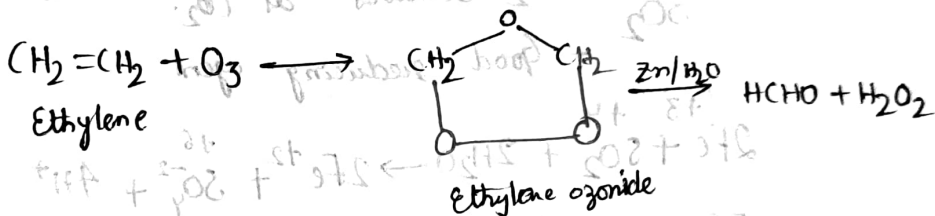
Bond angle 117°

Bond length 128 pm

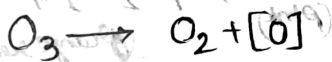
Bond order 1.5

Uses of Ozone:

- Germicide & bactericide
- In sterilization of water
- Atm. purification in mines, etc.
- O_3 is used for detection of unsaturation.



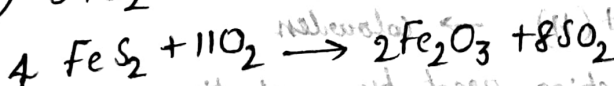
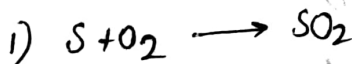
- Bleaching agent, oils, flour, starch.



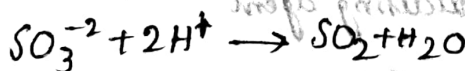
- $O_3 + (CN)_2$ — Rocket fuel
(cyanogen)

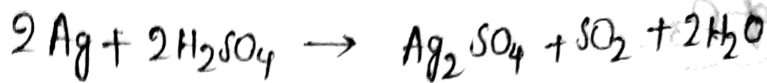
SO_2 : sulphur dioxide

"S" burn in O_2



- 2) In lab sulphite ion with del. Acids

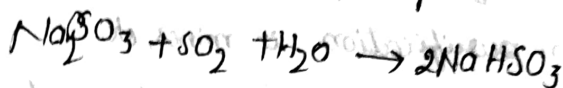
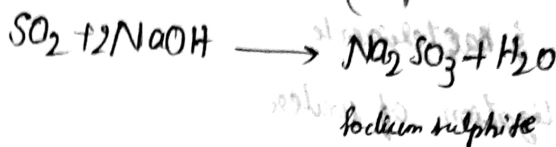
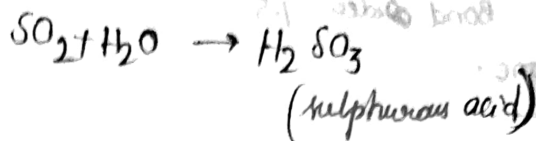




SO_2 - colourless gas

Choking smell

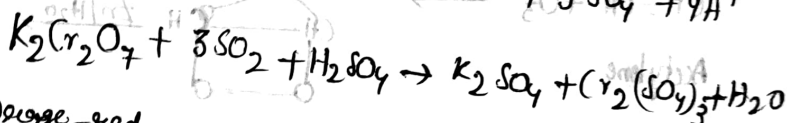
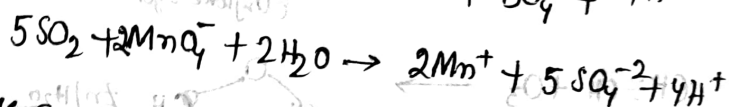
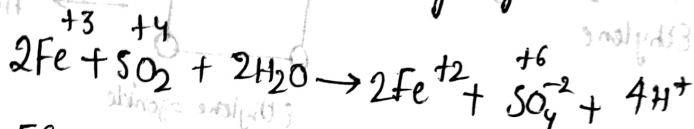
It is condensed easily by applying 2 atm pressure



Above reactions SO_2 behaves as CO_2 . sodium bisulphite

SO_2

Good reducing agent

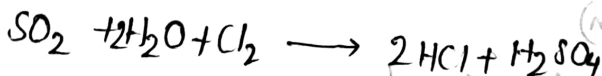


Orange-red

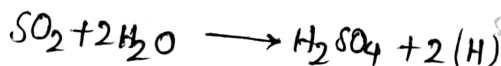
Green

→ SO_2 bleaches colour of KMnO_4 & $\text{K}_2\text{Cr}_2\text{O}_7$ by reduction.
(pink) (orange-red)

→ SO_2 reduces halogen into halogen acid.



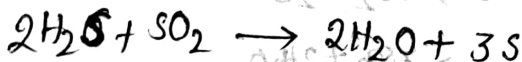
→ SO_2 - Bleaching agent

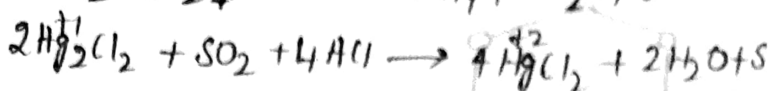


Coloured + (H) → colourless

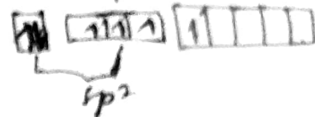
SO_2 acts as bleaching agent by reduction.

→ SO_2 acts as oxidising agent.





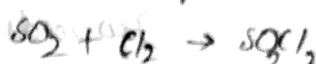
Text state - $3\text{s}^2 3\text{p}^3 3\text{d}^1$



one $\text{p} - \text{p} \pi$ bond

one $\text{d} \pi - \text{p} \pi$ bond

Angular



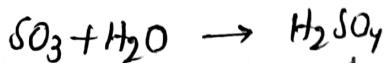
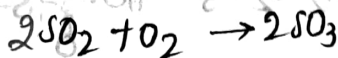
1) SO_2 used in refining of petrol and oil.

2) SO_2 used as antichlor in textiles.

3) SO_2 used as bleaching agent in wool & silk

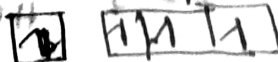
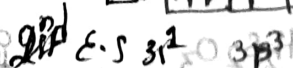
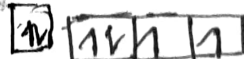
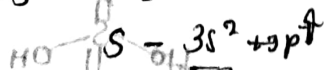
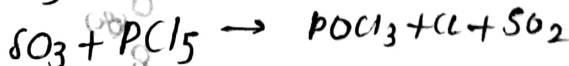
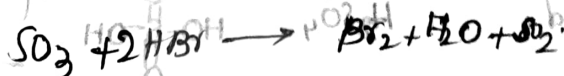
4) SO_2 used as preservative.

SO_3 (sulphur trioxide)



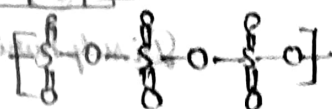
sulphuric acid

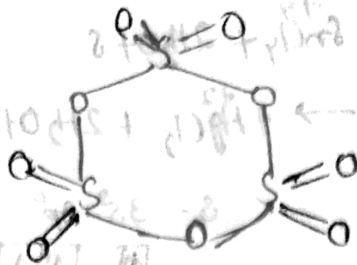
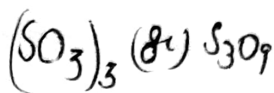
SO_3 - oxidising agent



α -form, β -form $\rightarrow$ stable

γ -form $\rightarrow$ cyclic trimer





No. of S-O bonds = 6

No. of S=O bonds = 3

Oxoacids of sulphur

I. Sulphurous acid series

	M.F	Structure	Te	C	Avg
1) Sulphurous acid	H_2SO_3		-	+4	4
2) Thion sulphurous acid	$H_2S_2O_3$		-2	+4	+1
3) Dithionous acid	$H_2S_2O_4$		+3	+3	+3
4) Pyrosulphurous acid	$H_2S_2O_5$		+3	+5	+4

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II. Sulphuric acid series

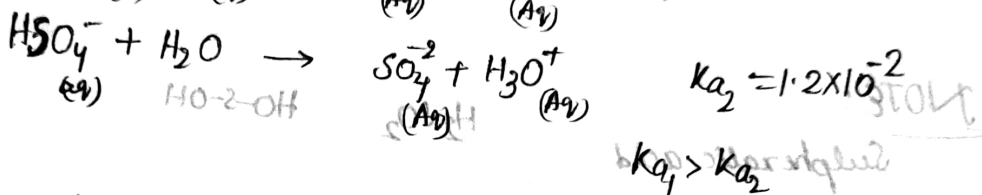
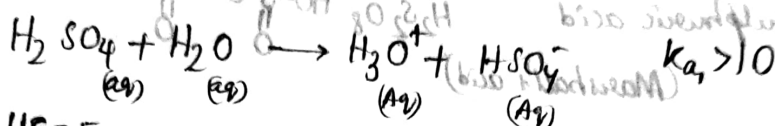
	M.F	Structure	T	C	Avg
1) Sulphuric acid	H_2SO_4		-	+6	+6
2) Thion sulphuric acid	$H_2S_2O_3$		-2	+6	+2
3) Pyrosulphuric acid (Oleum)	$H_2S_2O_7$		+6	+6	+6
Disulphuric acid					

→ When H_2O is added to it, it liberates heat

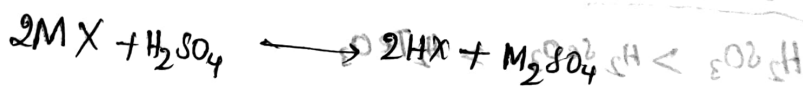
→ The chemical reactions of sulphuric acid is due to its

- low volatility
- strong acidic character
- acts as dehydrating agent
- oxidising agent

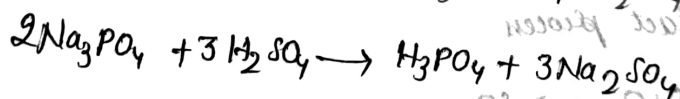
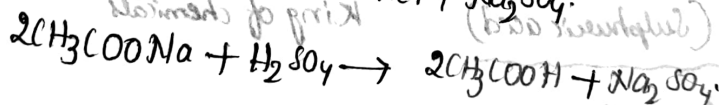
H_2SO_4 - strong acid - dibasic acid



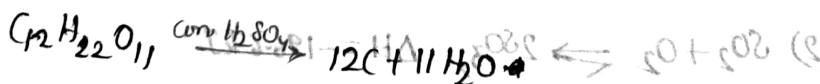
Low volatility



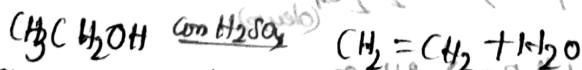
Here $X = F, Cl, NO_3$



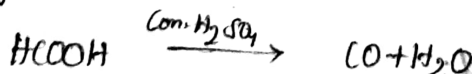
H_2SO_4 - dehydrating agent



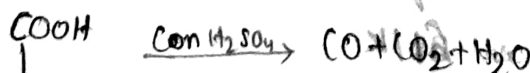
sucrose



Ethyl alcohol

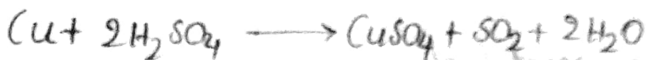


Formic acid



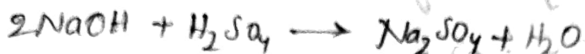
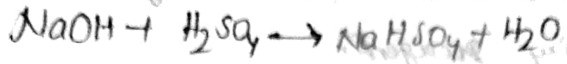
Oxalic acid

H_2SO_4 - Oxidising agent

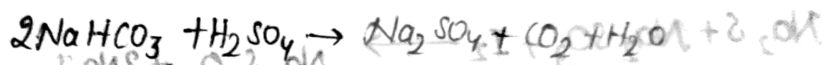
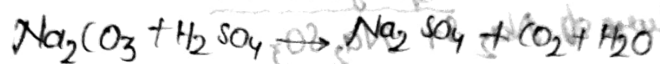


Carbon forms two acidic oxides (CO_2, SO_2) when it is oxidized with conc H_2SO_4 .

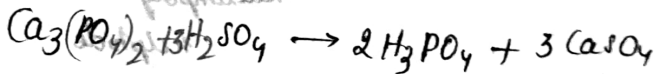
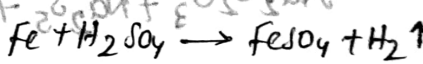
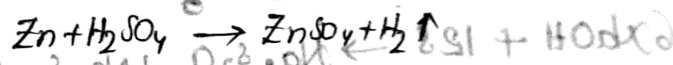
Acid



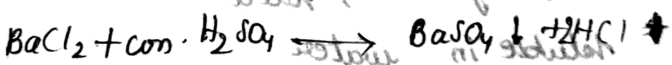
→ decomposes carbonates and bicarbonates and liberates CO_2 .



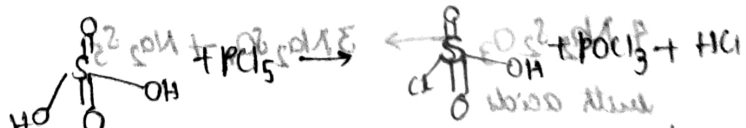
→ Reacts with metals liberates H_2



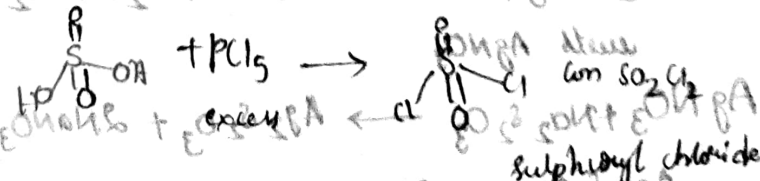
→ ppt reaction



→ Reaction with PCl_5



chlorosulphonic acid



sulphonyl chloride

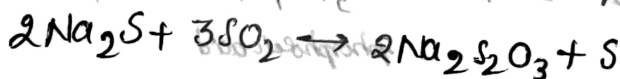
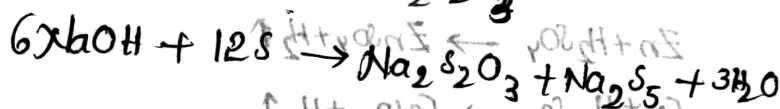
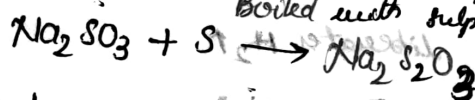
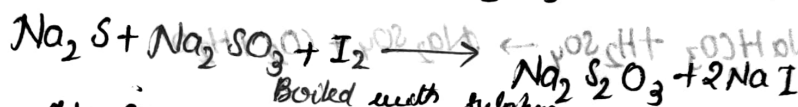
Uses of H_2SO_4

- Used as pickling agent (cleaning of metal surface)
- Used in manufacture of fertilizers, paints & pigments.
- Used as a nitration mixture along with conc. HNO_3 .
- Used in leather industry.
- Used in petrol refining process.
- Lead storage batteries & acid storage batteries.
- Used as laboratory reagent.

24/4/23

Sodium thiosulphate ($Na_2S_2O_3$)

"Hypo" mixture of Na_2S + Na_2SO_3 Sprungs method

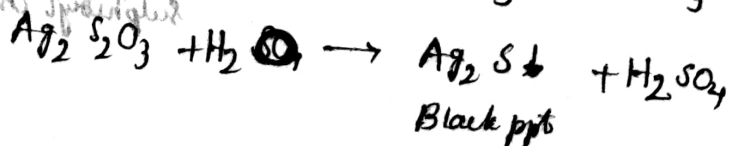
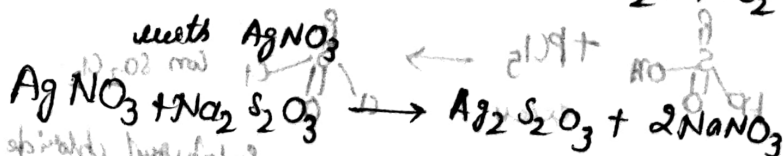
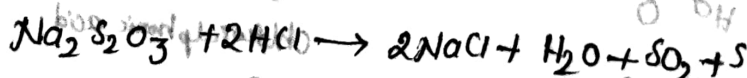


Hypo - Colourless, crystalline, solid

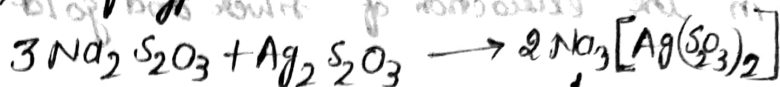
soluble in water

Hypo is stable upto 488K

at 498K

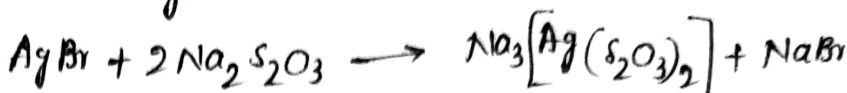


Excess of Hypo



↓
Sodium argento thiosulphate

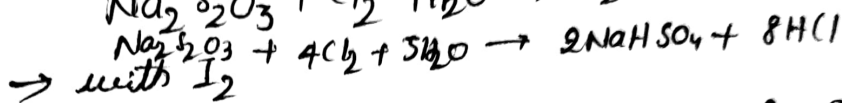
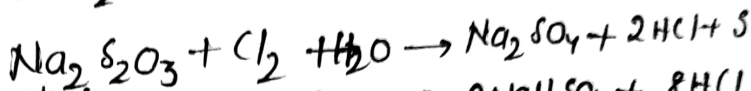
→ with AgBr



above reaction is useful in photography

Hypo is used as fixing agent in photography

→ with Cl_2



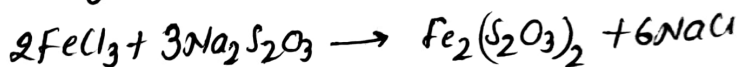
→ with I_2



Sodium tetrathionate

useful in iodometric and iodometric titrations

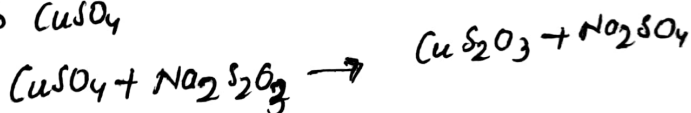
→ with FeCl_3



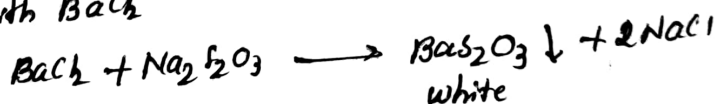
Excess

violet

→ with CuSO_4

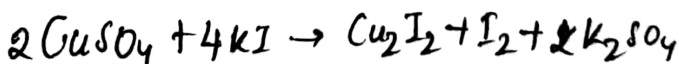
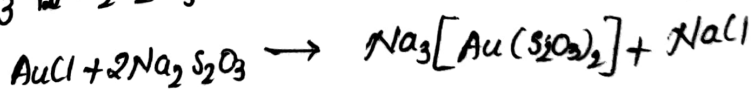
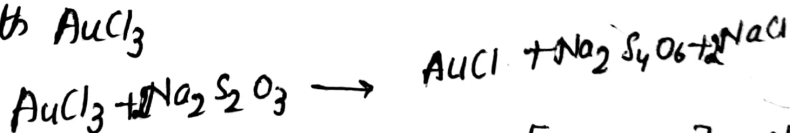


→ with BaCl_2



white

→ with AuCl_3



Uses

→ As a antichlor agent

→ As a fixing agent in photography

→ Used as a iodometric and iodometric titrations

→ Used as antiseptic.